

Warm-up

- (1) Give a parameterization of a curve that goes clockwise around an ellipse with horizontal minor radius 1 and vertical major radius 3.
- (2) Set up an integral to compute the length of the curve given parametrically by $r(t) = (\sqrt{t}, 6 \sin(t))$ from $t = 0$ to $t = 2\pi$.

Medium

- (1) Find a (non-parametric) equation of the line tangent to the curve parameterized by $x = \cos^3(t)$, $y = \sin^3(t)$ at the point $(1/8, -3\sqrt{3}/8)$.
- (2) Find the length of the curve parameterized by $x = e^t - t$, $y = 4e^{t/2}$ for $0 \leq t \leq 2$.
- (3) Find a parameterization of the ellipse $x^2/4 + y^2 = 1$.
- (4) Use calculus to find the exact coordinates of the leftmost point on the parametric curve $x = t^4 - t^2$, $y = t + \ln(t)$.
- (5) Find the length of the curve parameterized by $x = e^t \cos(t)$, $y = e^t \sin(t)$.

Stretch

- (1) A *cycloid* is a curve traced out by a point on the circumference of a circle of radius r rolling along a straight line. Supposing the point begins at $(0, 0)$ and the circle rolls along the positive x -axis in the xy -plane, find parametric equations for the cycloid.
- (2) Use your answer in (1) to find the length of one arch of the cycloid.